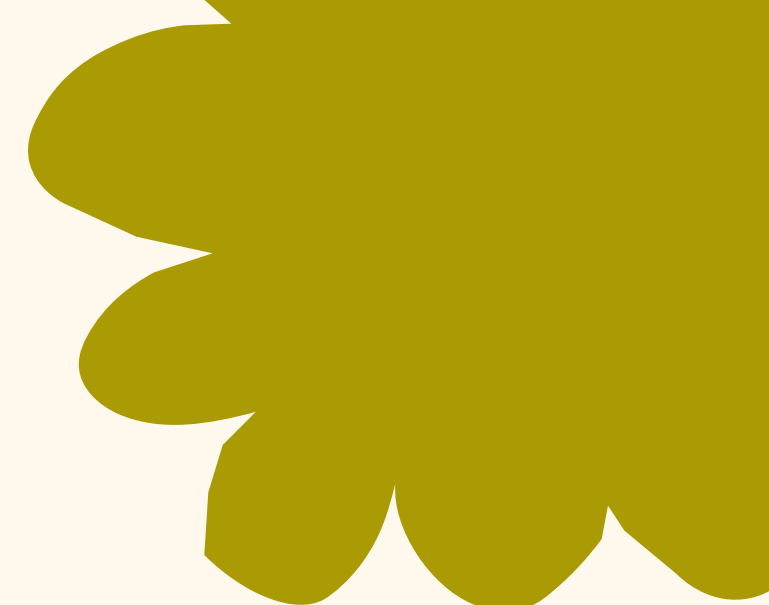
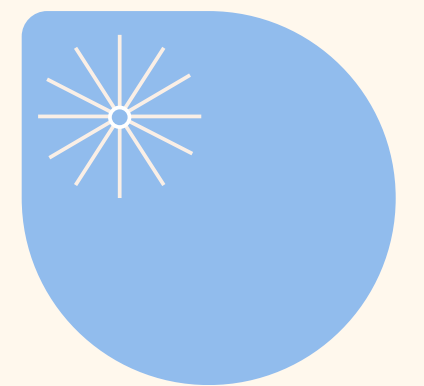
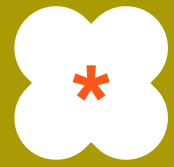




Exploring Neighborhood Crime, Equity, and After- School Programs

Team Members:
Moose, Stuti and Sam





Problem Definition & Research Question:

Research Question: What are the relationships between neighborhood crime rates, equity scores, demographics and after-school programs?

Real-World Impact: Evidence-based policy reform for 78 Denver neighborhoods

Why This Matters:

- Informs \$500K+ annual program investments
- Challenges current crime prevention strategies
- First analysis linking Denver's equity data with crime patterns

Dataset Description & Metadata

Denver Open Data Catalog

Four Comprehensive Data Sources:

Crime Data: 50,000+ incidents → 78 neighborhoods

- Metadata: Geographic coordinates, incident types, timestamps
- Processing: Calculated crime rates per 1,000 residents

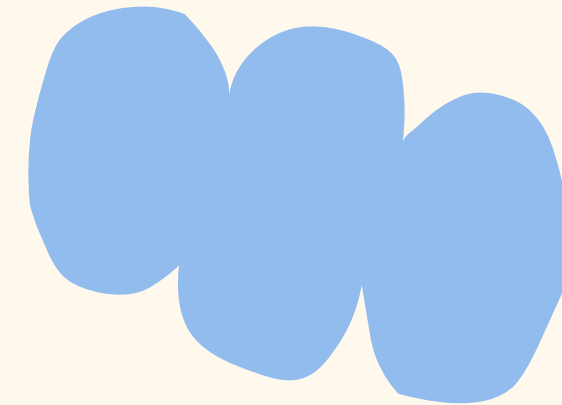
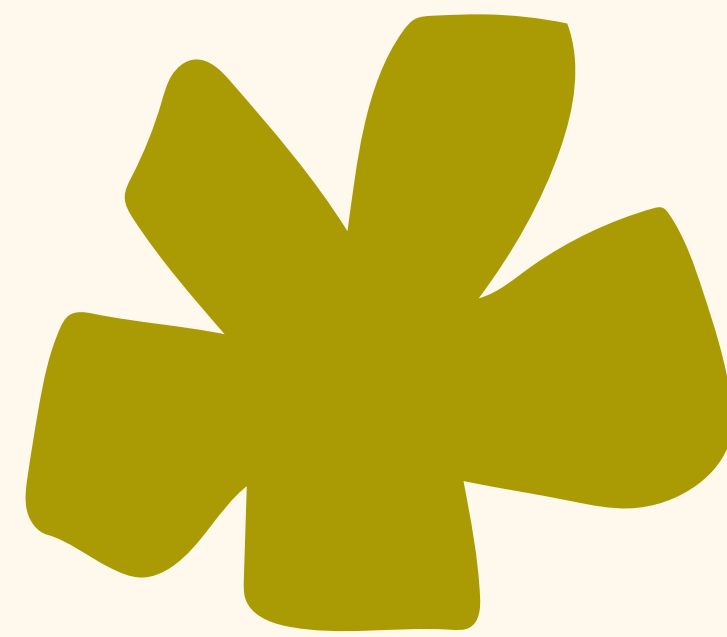
After-School Programs: 200+ locations mapped

- Metadata: Program capacity, age groups, locations
- Processing: Program density per 1,000 residents

Equity Scores: Denver's official 1-5 scale

- Metadata: Income, education, health access composite

Demographics: Census tract population data





Data Cleaning & Feature Engineering

Major Technical Challenges Solved

Geographic Standardization:

- Mapped 47,892 crime incidents to neighborhood boundaries
- Resolved coordinate system inconsistencies across datasets

Rate Calculations:

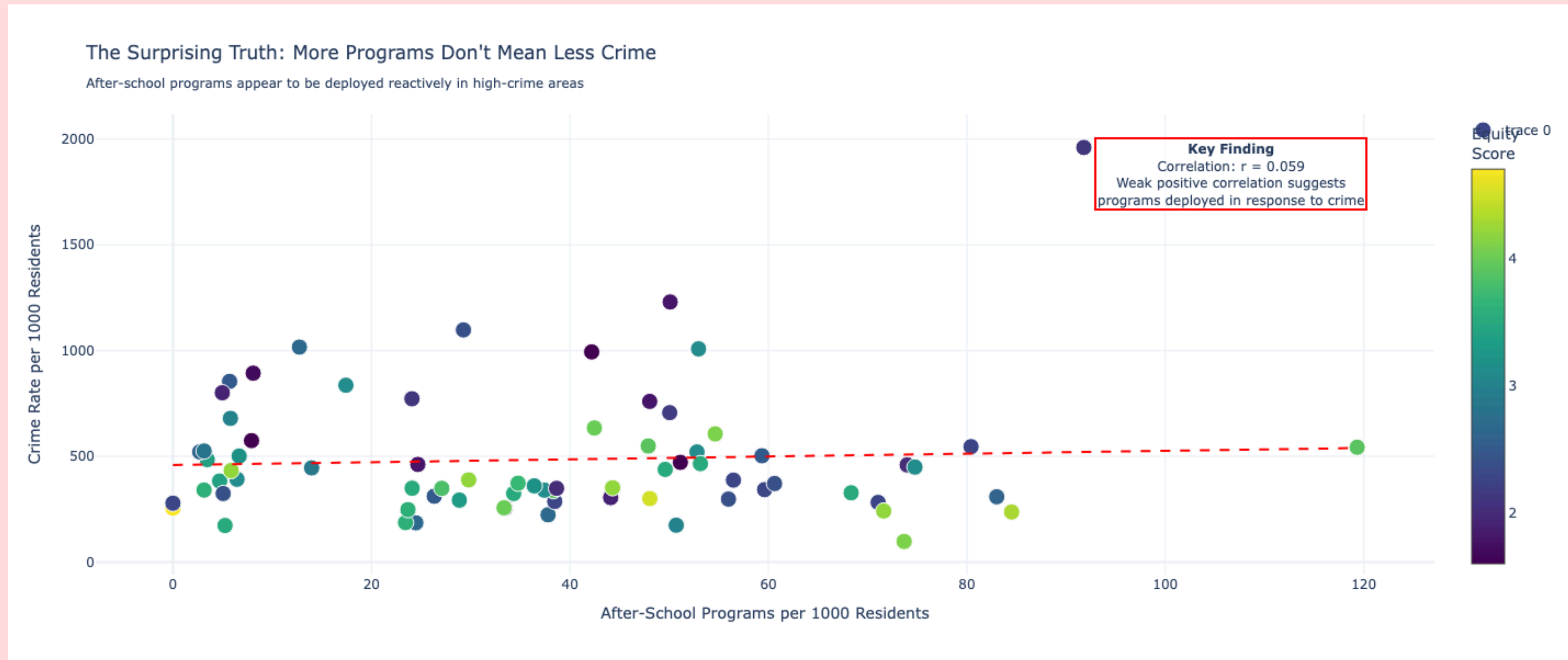
- Created fair comparison metrics (per 1,000 residents)
- Normalized program density by population

Quality Control:

- Validated 78 residential neighborhood assignments
- Cross-referenced program locations with city records

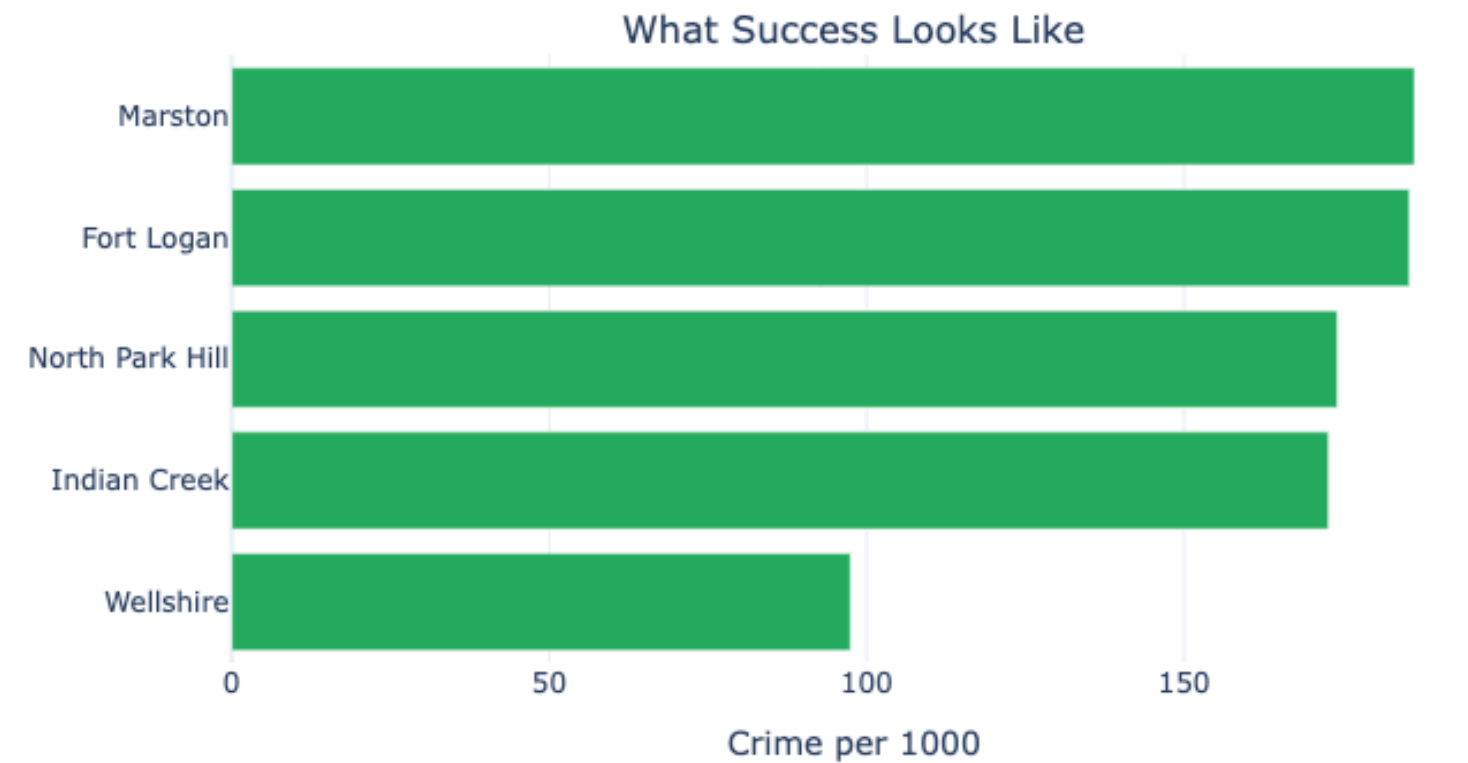
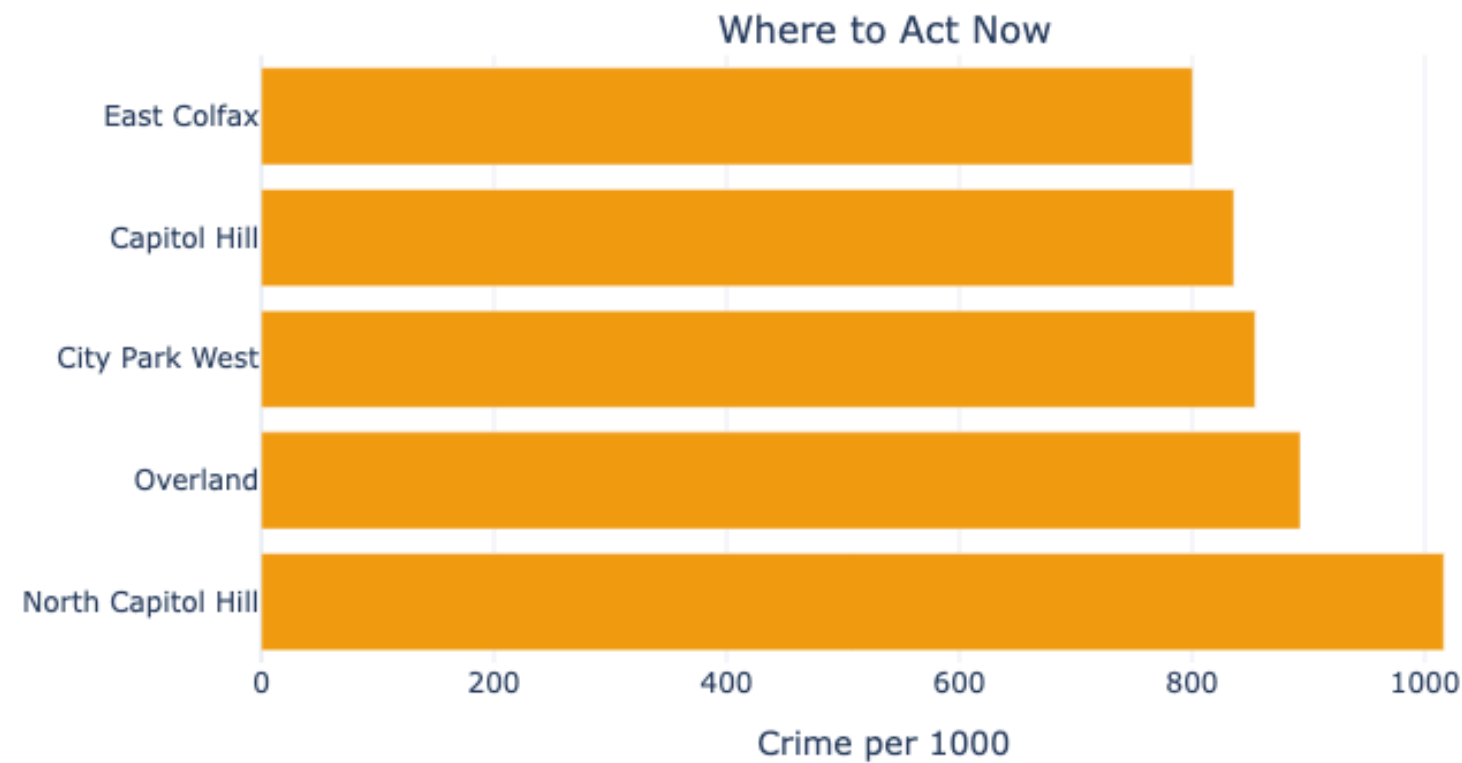
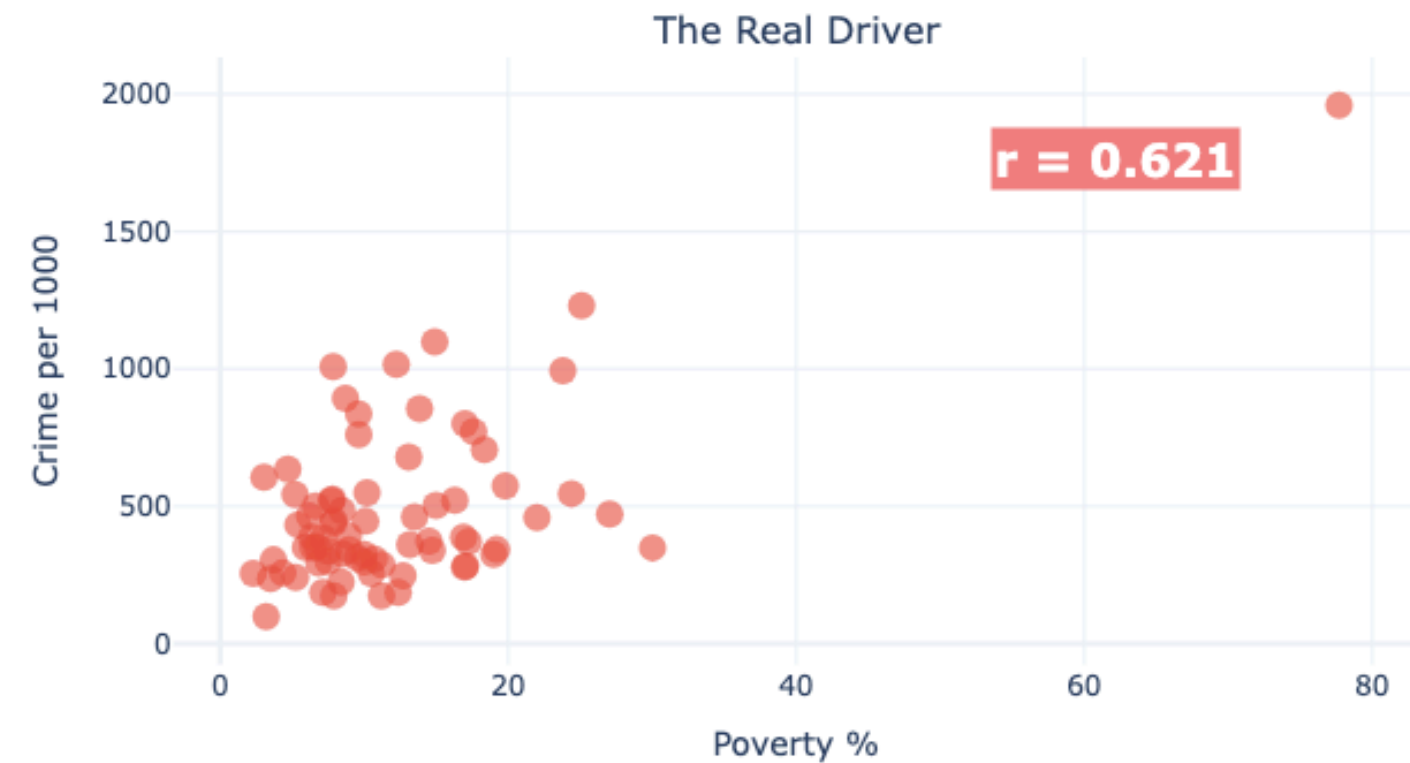
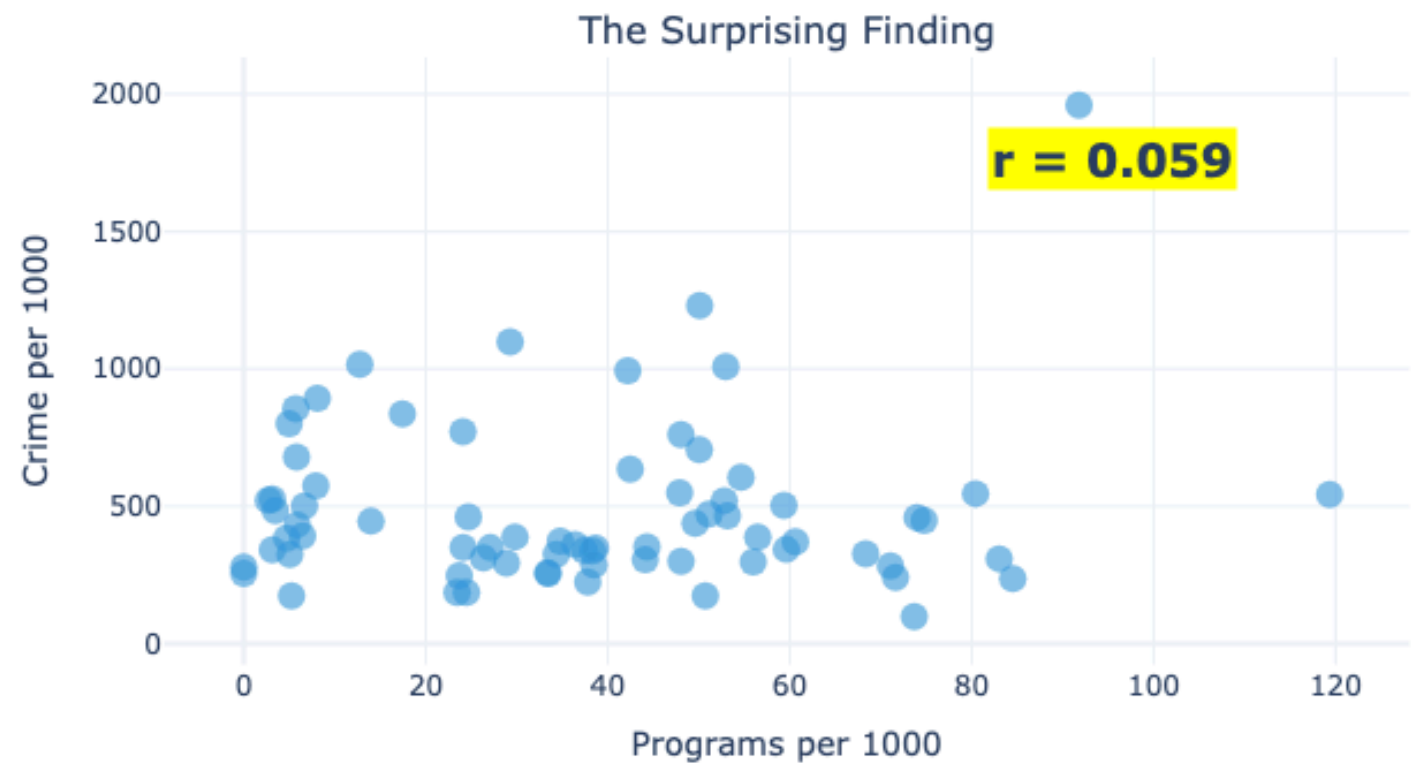


The Shocking Discovery: Programs Follow Crime

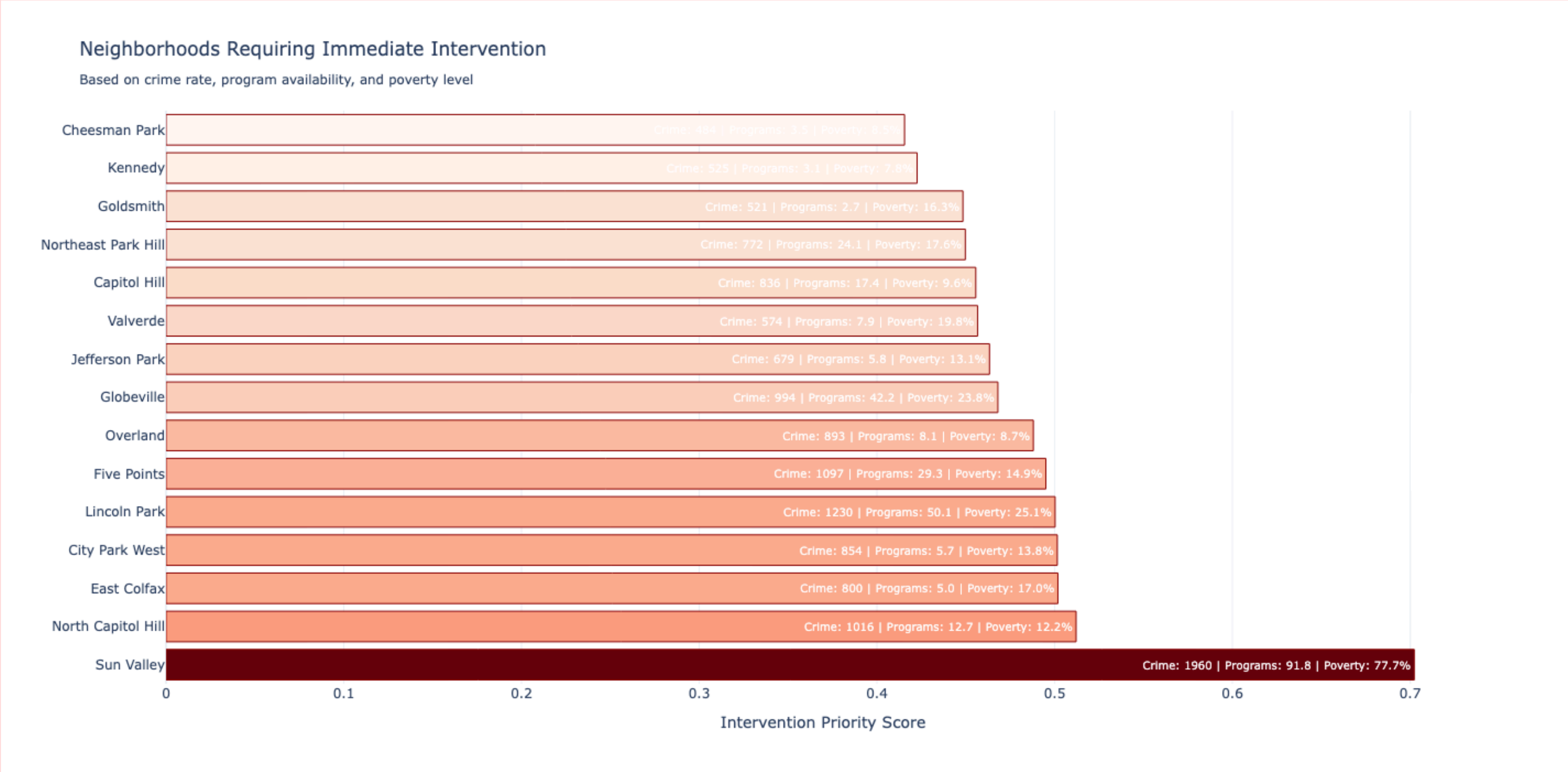


Four Key Insights Drive Policy Change

Denver Crime Analysis: Key Insights at a Glance



15 Neighborhoods Need Immediate Action



Analysis Results & Interpretation

Statistical Relationships Revealed

Programs → Crime: $r = +0.069$ (weak positive) Reactive deployment - programs follow crime, don't prevent it

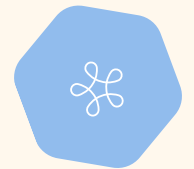
Poverty → Crime: $r = +0.621$ (moderate positive)
Socioeconomic factors drive crime more than programs

Education → Crime: $r = -0.036$ (weak negative) Limited impact alone - comprehensive approach needed

Key Discovery: 15 high-priority neighborhoods identified Sun Valley leads intervention priority (score: 0.77)

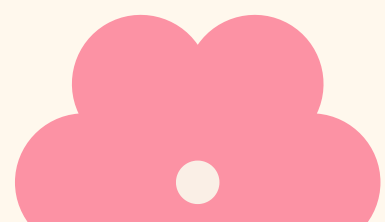
Statistical Significance: All correlations $p < 0.05$

Classification and Regression Models

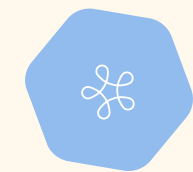


These models were created to help answer subquestions within our overall exploration of the relationship between neighborhood crime, equity, school programs, and selected demographics.

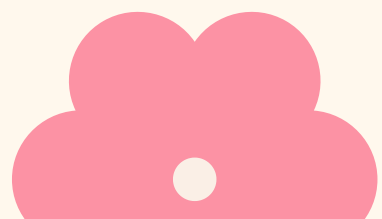
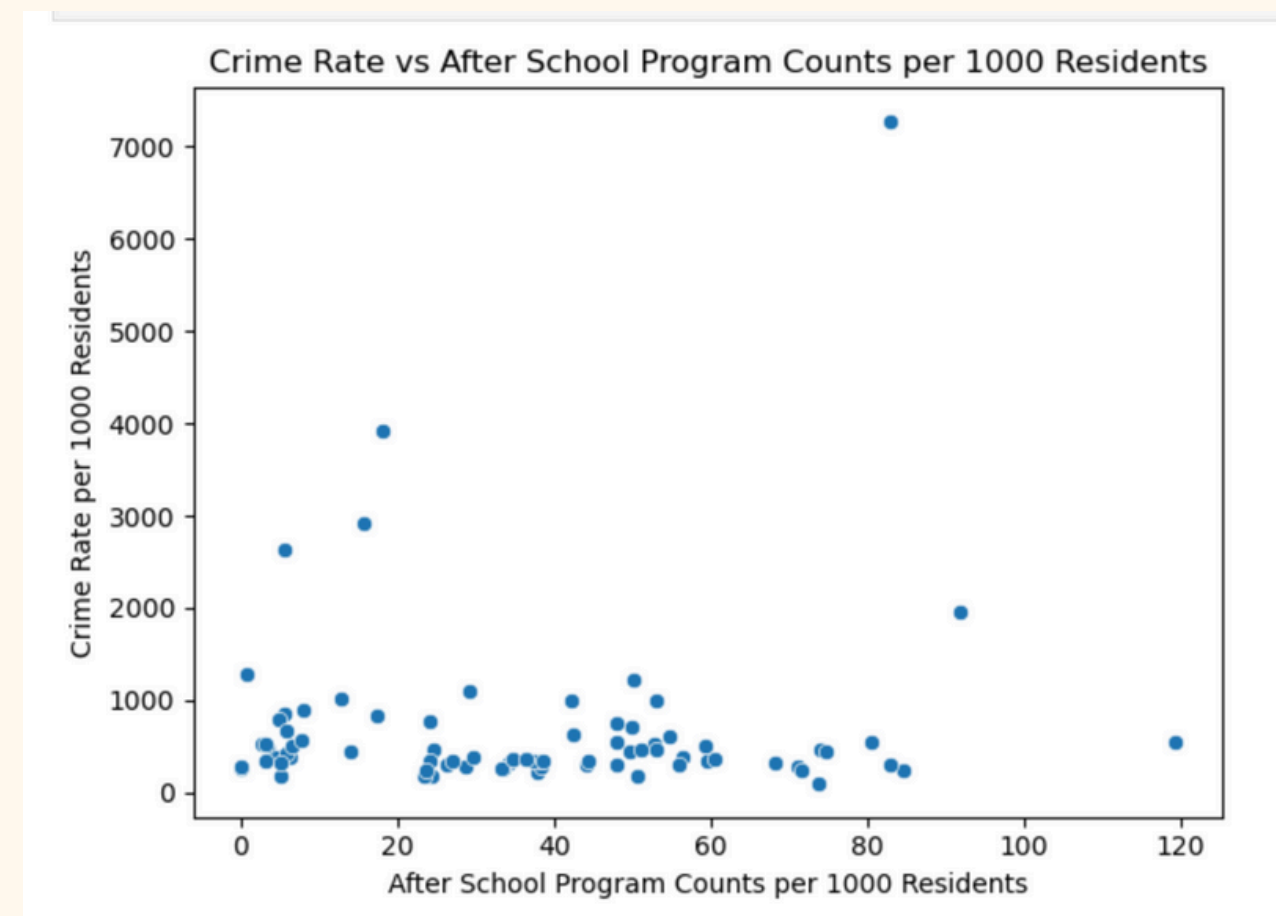
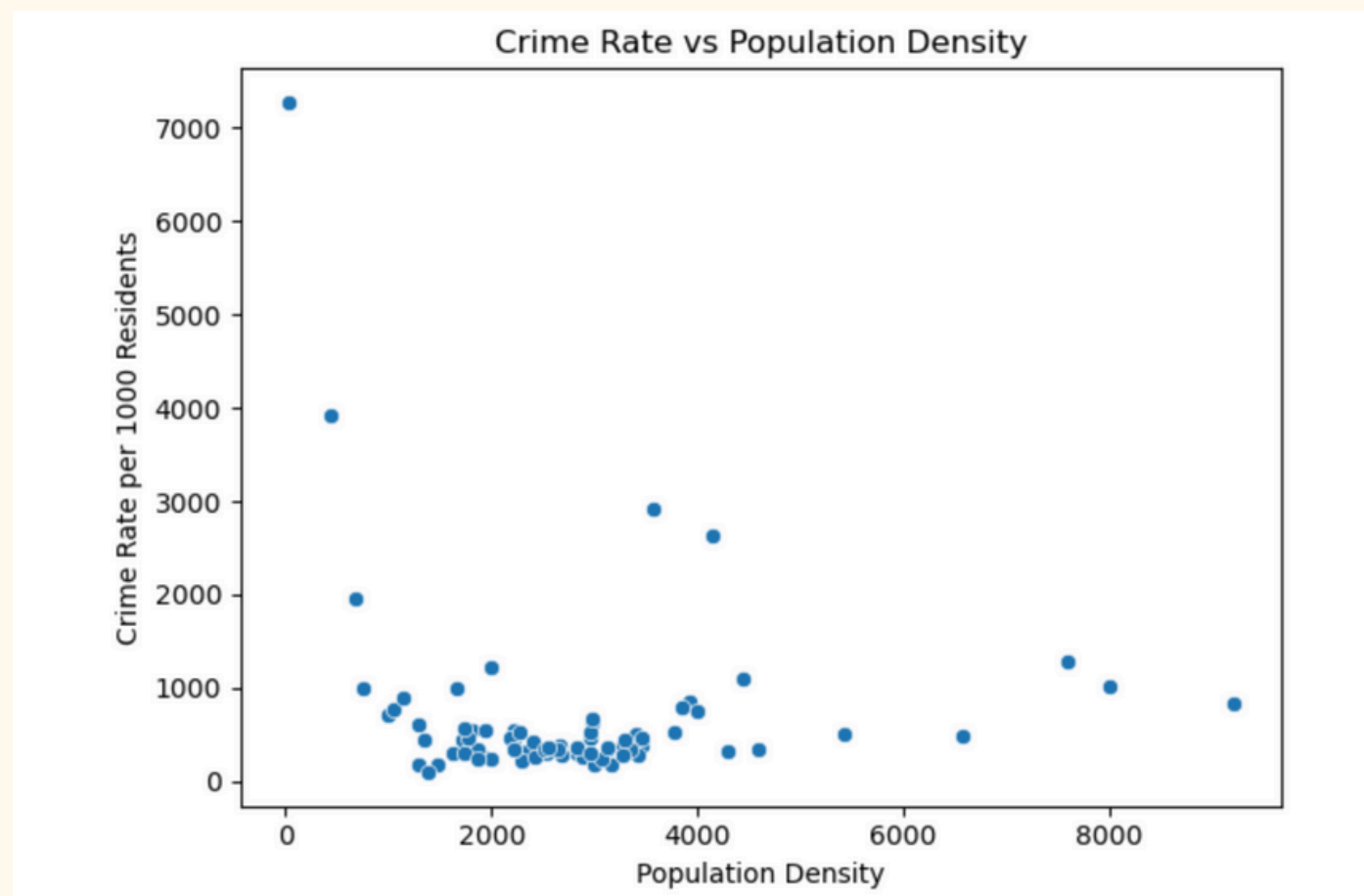
- Can we predict the neighborhood crime rate given other features?
- Which inputs carry the most importance for neighborhood crime rate?
- Can we predict neighborhood equity scores given other features?
- What inputs carry the most importance for different equity score tiers and continuous average equity scores?



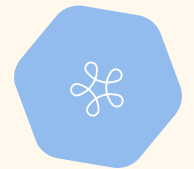
Crime Linear Regression - Setup



After merging data into a single data frame, further EDA was conducted to find any possible linear relationships between feature columns and the output, crimes per 1000 residents. These plots did not reveal any obvious connections between neighborhood-level data and crime.



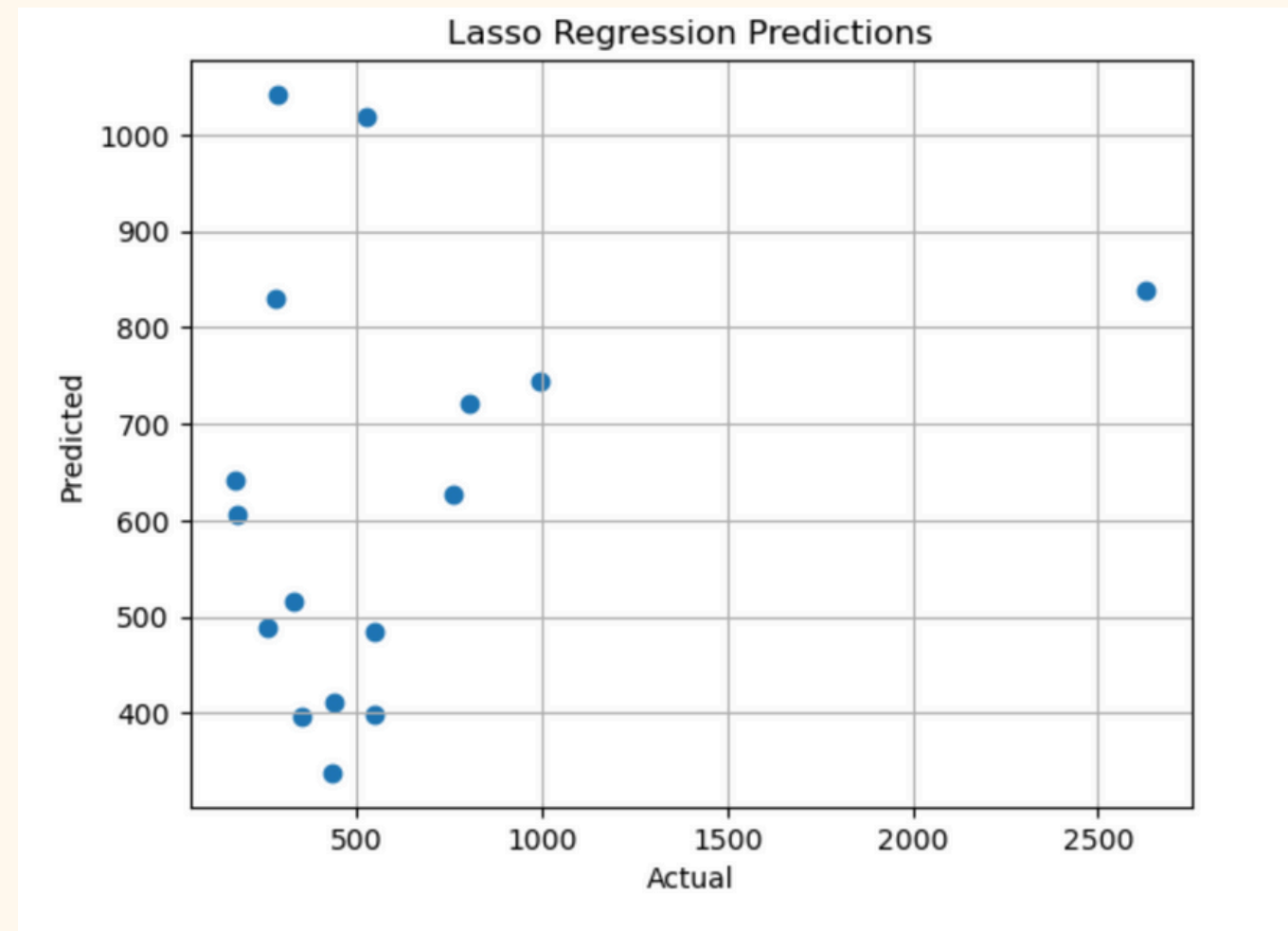
Crime Linear Regression - Results



Two predictive models were created, one with Lasso l1 regularization and one with no regularization. Both had very low R2 values, and very high RMSE values, leading to the conclusion that the neighborhood data did not provide a clear picture into what features are connected to higher crime rates, and how crime rates can be predicted.

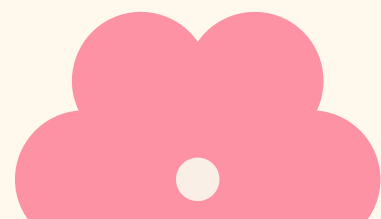
Lasso Model

R2 Score: 0.05400428755936881
RMSE: 555.8155722928764

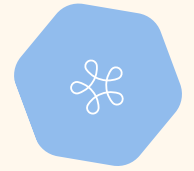


No Regularization Model

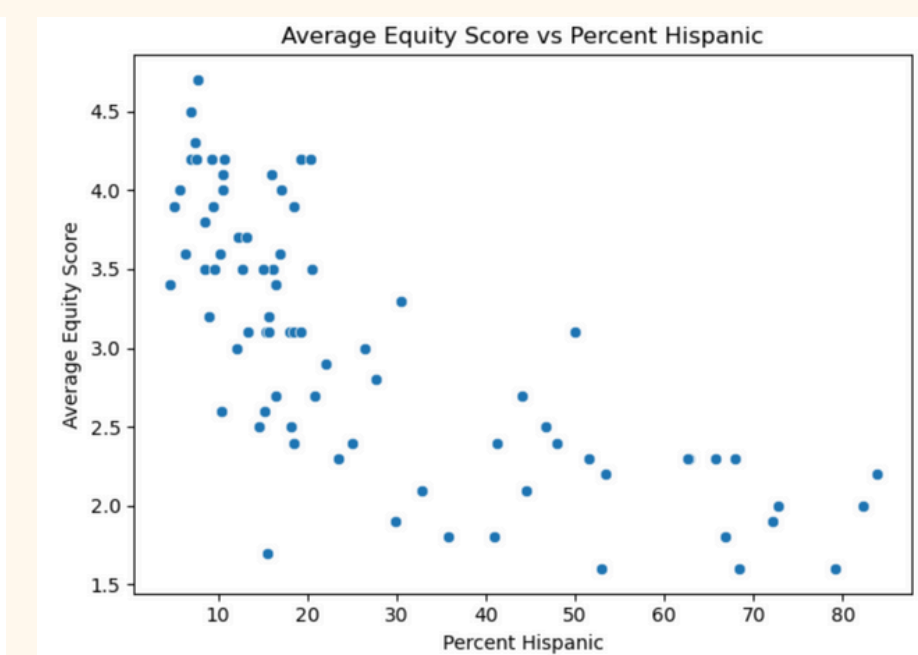
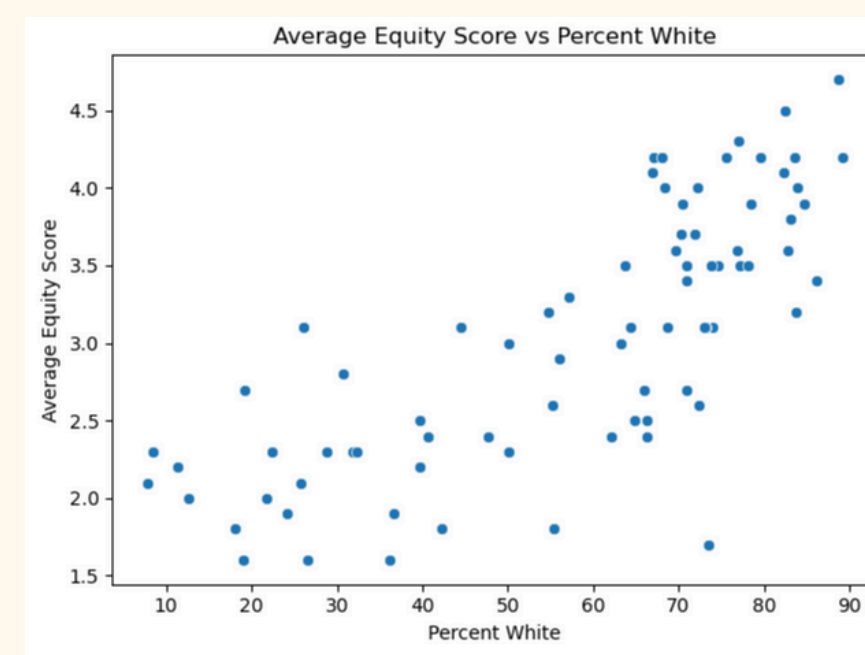
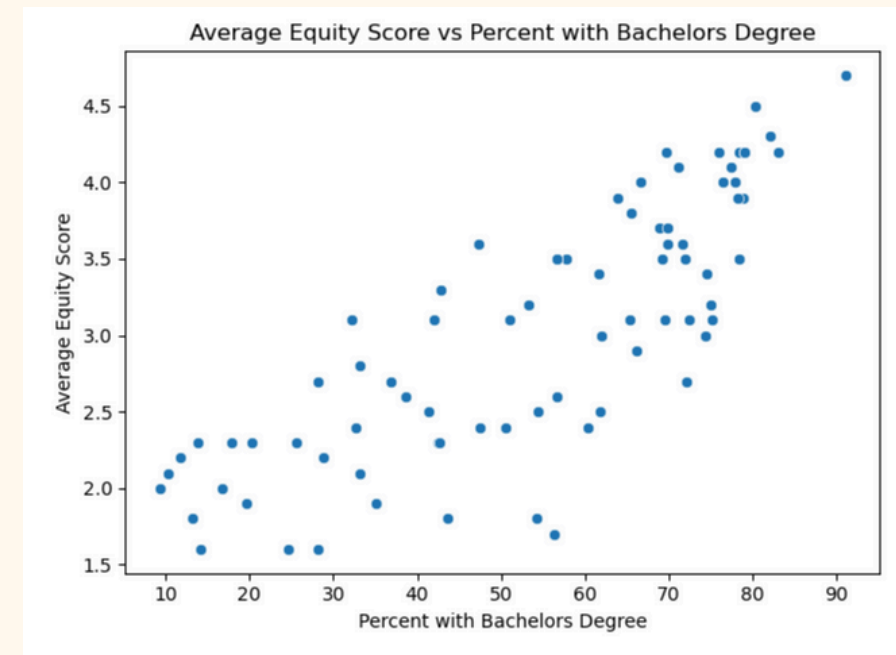
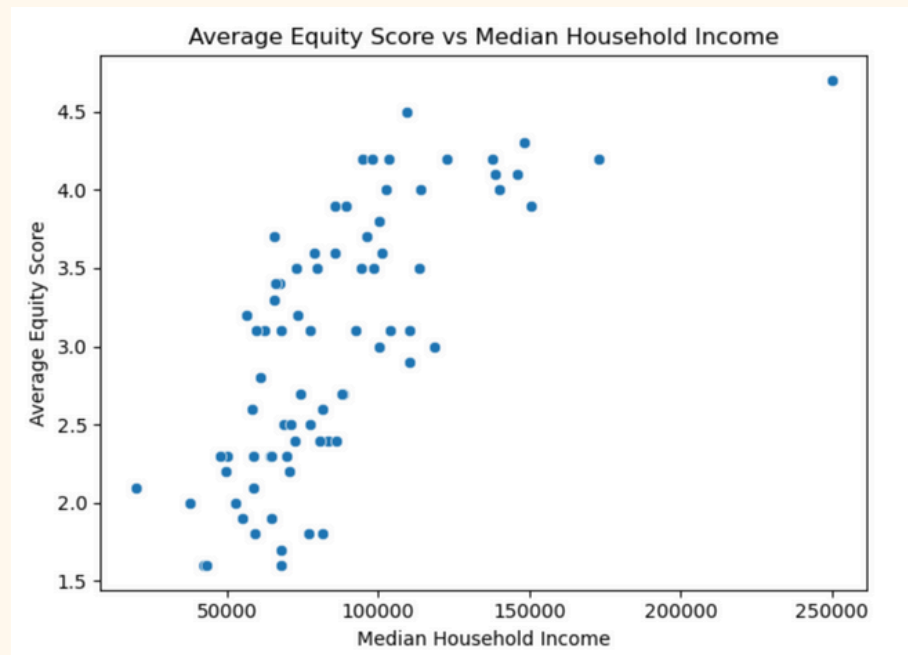
R2 Score: 0.04907409523965589
RMSE: 557.2620463971798



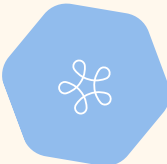
Equity Linear Regression - Setup



After plotting different demographics data to the average equity values per neighborhood, it was determined that there were some clear linear relationships to explore.



Equity Linear Regression - Results

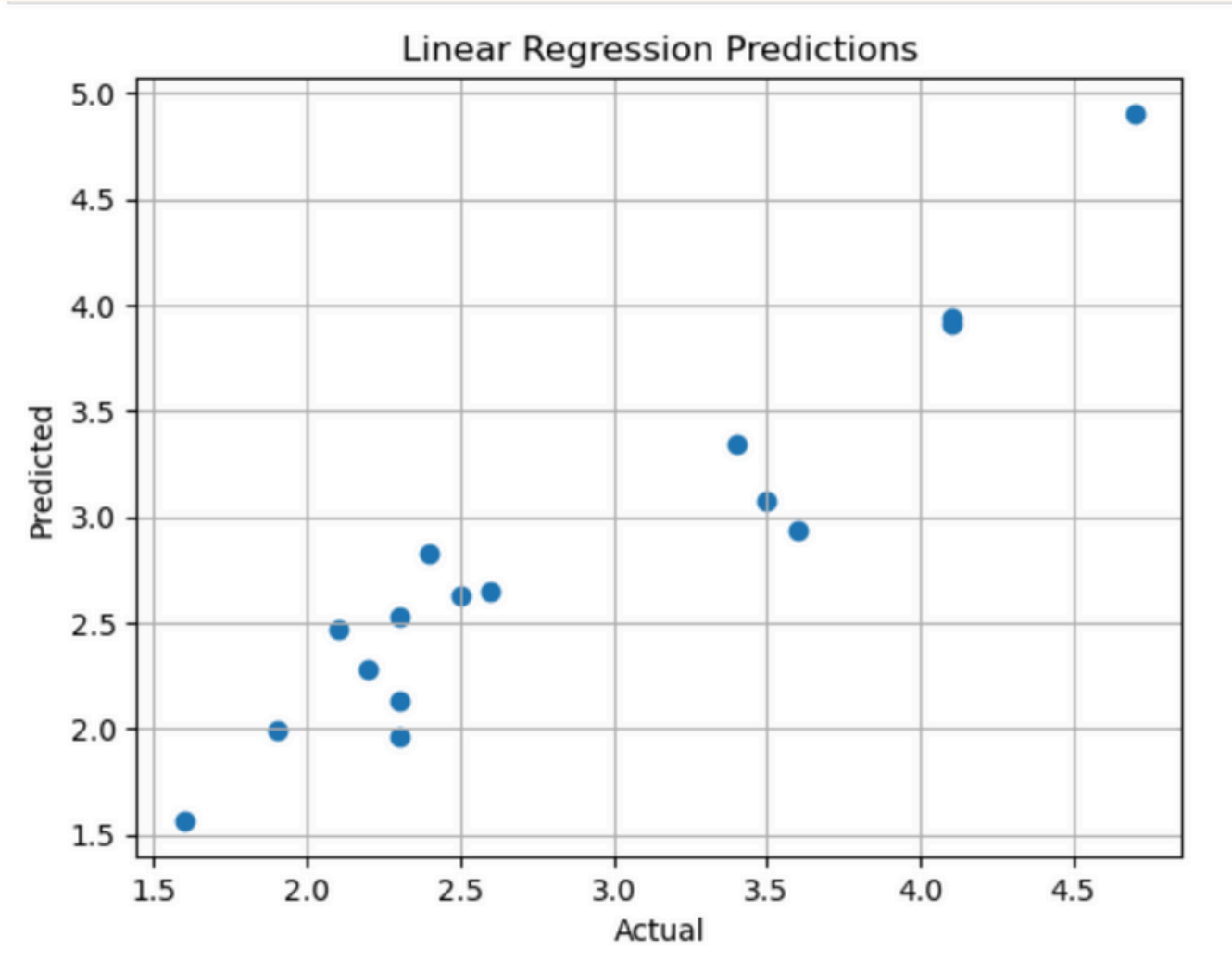


It was clear the set of features can be used to explain the variance in the average equity score values per neighborhood, and should be included in any predictive modeling.

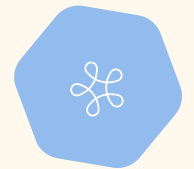
R2 Score: 0.8986948401764887
RMSE: 0.28222539803274804

OLS Regression Results						
=====						
Dep. Variable:	y	R-squared:	0.721			
Model:	OLS	Adj. R-squared:	0.693			
Method:	Least Squares	F-statistic:	25.84			
Date:	Fri, 30 May 2025	Prob (F-statistic):	4.42e-17			
Time:	19:47:43	Log-Likelihood:	-45.831			
No. Observations:	78	AIC:	107.7			
Df Residuals:	70	BIC:	126.5			
Df Model:	7					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	3.0256	0.052	58.133	0.000	2.922	3.129
Median Household Income	0.2249	0.081	2.768	0.007	0.063	0.387
program_count_per1000	0.0565	0.057	0.991	0.325	-0.057	0.170
percent_bachelors	0.2825	0.169	1.667	0.100	-0.055	0.620
Percent White	0.0463	0.160	0.290	0.773	-0.273	0.365
Percent Living in Poverty	-0.0853	0.075	-1.139	0.258	-0.235	0.064
percent_less_hs	0.1005	0.202	0.497	0.621	-0.303	0.504
Percent Hispanic	-0.2724	0.185	-1.476	0.144	-0.640	0.096
=====						
Omnibus:	4.096	Durbin-Watson:	2.263			
Prob(Omnibus):	0.129	Jarque-Bera (JB):	3.611			
Skew:	-0.524	Prob(JB):	0.164			
Kurtosis:	3.104	Cond. No.	10.8			
=====						



Equity Logistic Regression - Setup



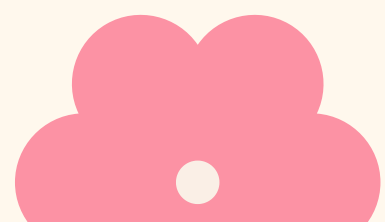
Our original proposal included creating tiers for the average equity scores to be used as the labels in a logistic regression task. The equity scores were rounded to the nearest integer value as the label, and the single neighborhood with a score of 5 included in the class 4 for simplification.

Output Class Labels

```
Class 2: 28 instances  
Class 3: 21 instances  
Class 4: 29 instances
```

Stratified data split

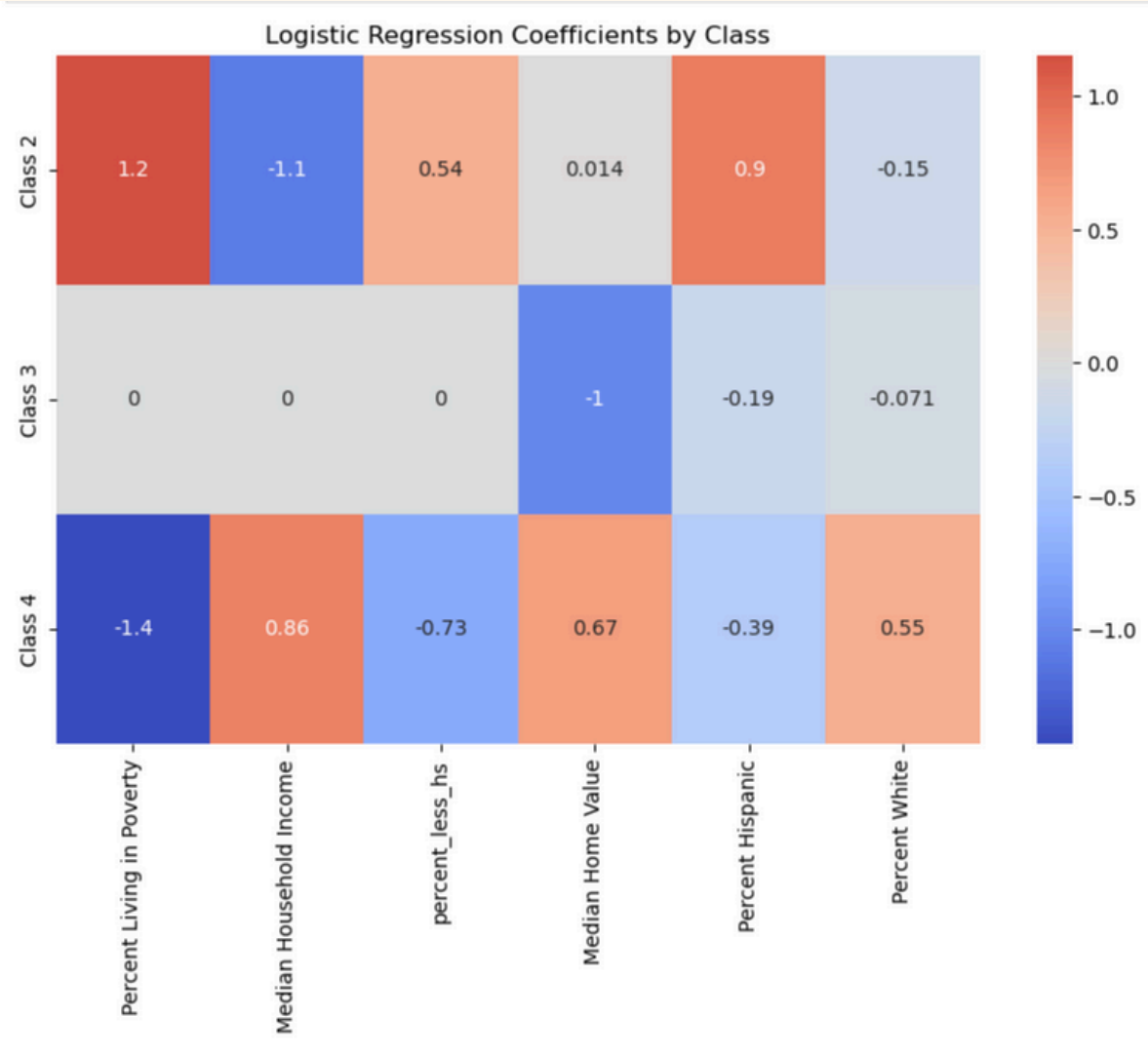
```
Train distribution: {2: 22, 3: 17, 4: 23}  
Test distribution: {2: 6, 3: 4, 4: 6}
```



Equity Logistic Regression - Results

After creating multiple models, using regularization and hyperparameter tuning with GridSearchCV, a final elasticnet model was created.

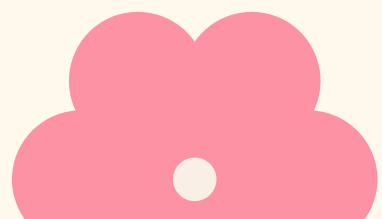
	Percent Living in Poverty	Median Household Income	percent_less_hs	Median Home Value	Percent Hispanic	Percent White
Class 2	1.154009	-1.091405	0.544076	0.013810	0.900847	-0.153247
Class 3	0.000000	0.000000	0.000000	-1.007419	-0.188604	-0.071386
Class 4	-1.427850	0.855103	-0.729878	0.668774	-0.385029	0.552005



Classification and Regression Models - Conclusion

We were able to uncover some meaningful insights using these models, mostly in working towards answering the questions relating to which features are most important in predicting equity scores and those that can be attributed to the variance in the output.

One issue across all models was the limited dataset. With only 78 official neighborhoods, these models did not receive a large amount of training data to use.

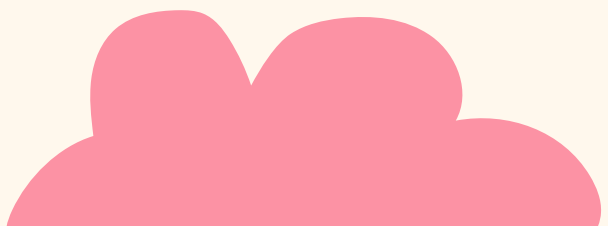


Time Series Analysis of Denver Crime Data

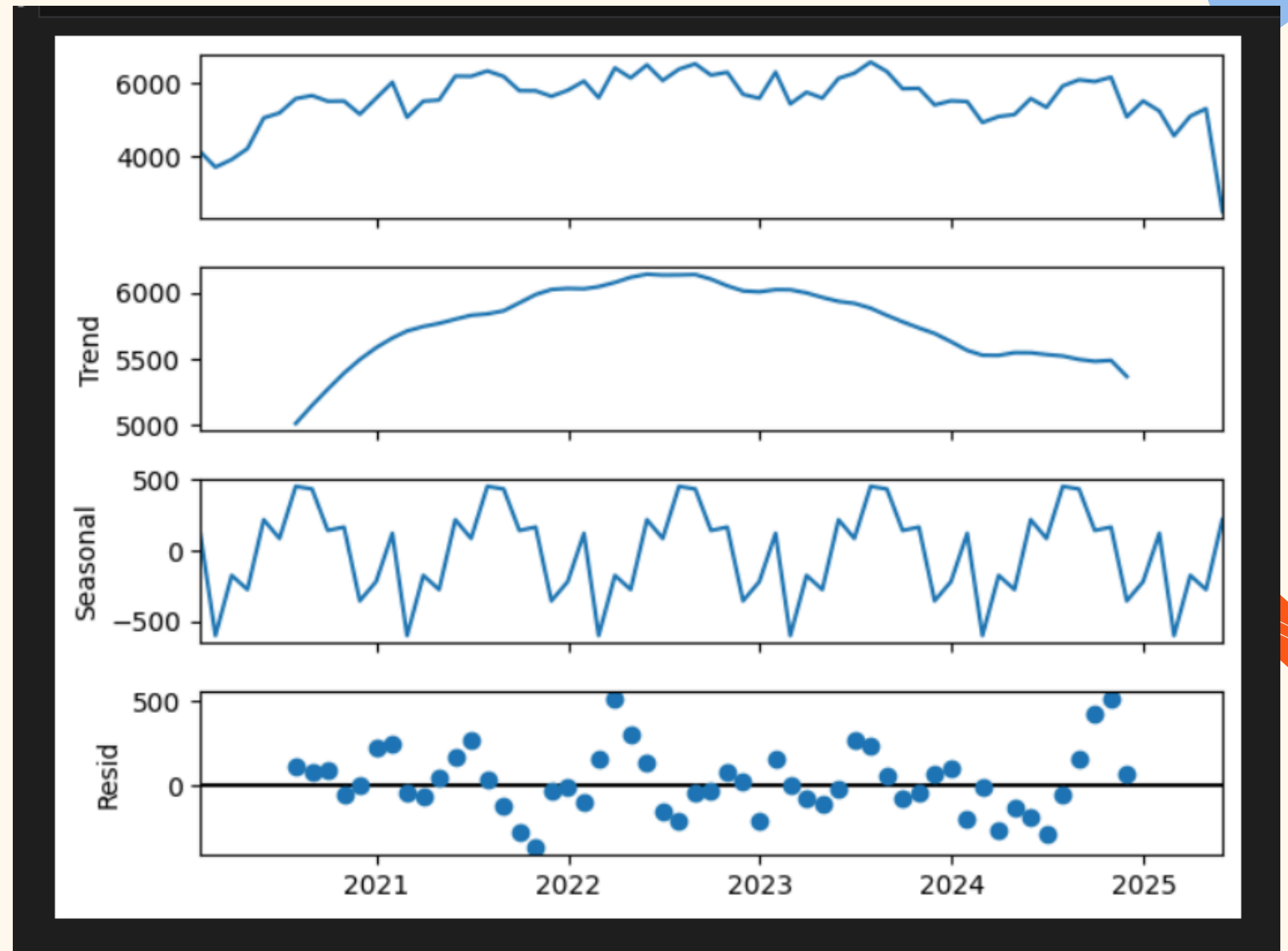


Why analyze crime data?

- Public safety planning and resource allocation
- Anticipating future trends for proactive measures Goals
- Explore historical crime trends in Denver
- Decompose time series to understand seasonality and trends
- Forecast future crime rates using Prophet



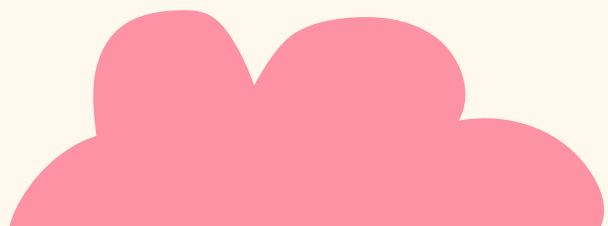
- Used seasonal_decompose (period=12 months)
- Components:
 - Trend: Long-term direction of crime rates
 - Seasonality: Recurring monthly patterns
 - Residual: Unexplained noise
- Key Insights:
 - Crime rates tend to peak in summer months and dip in winter



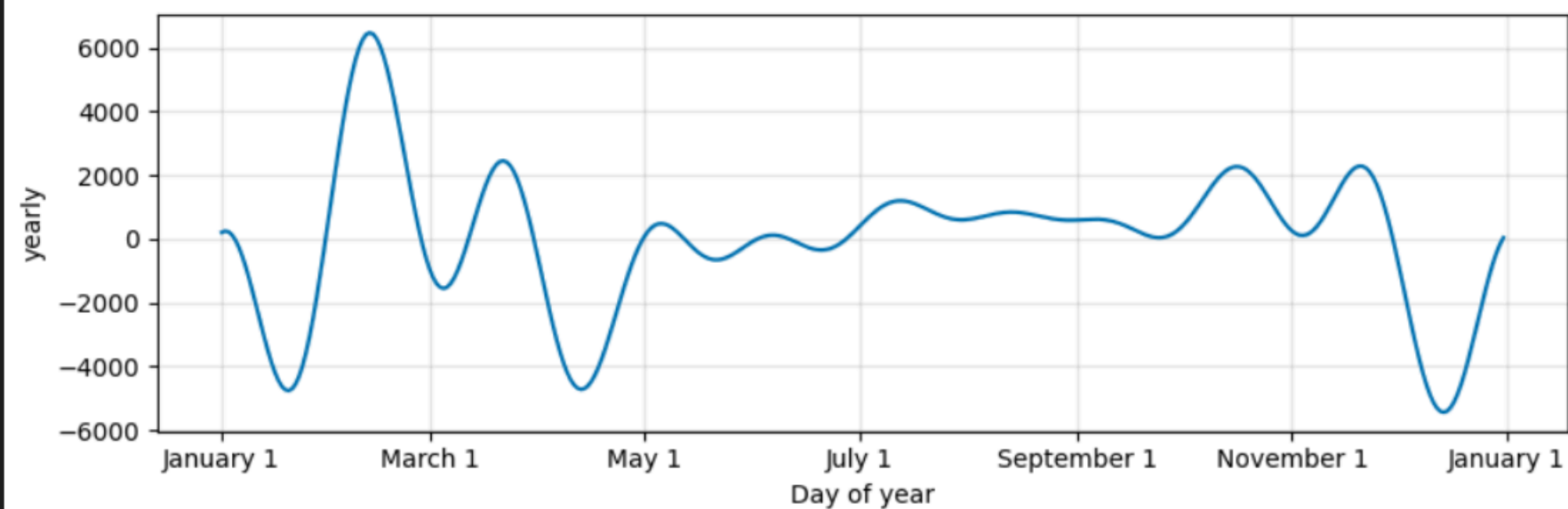
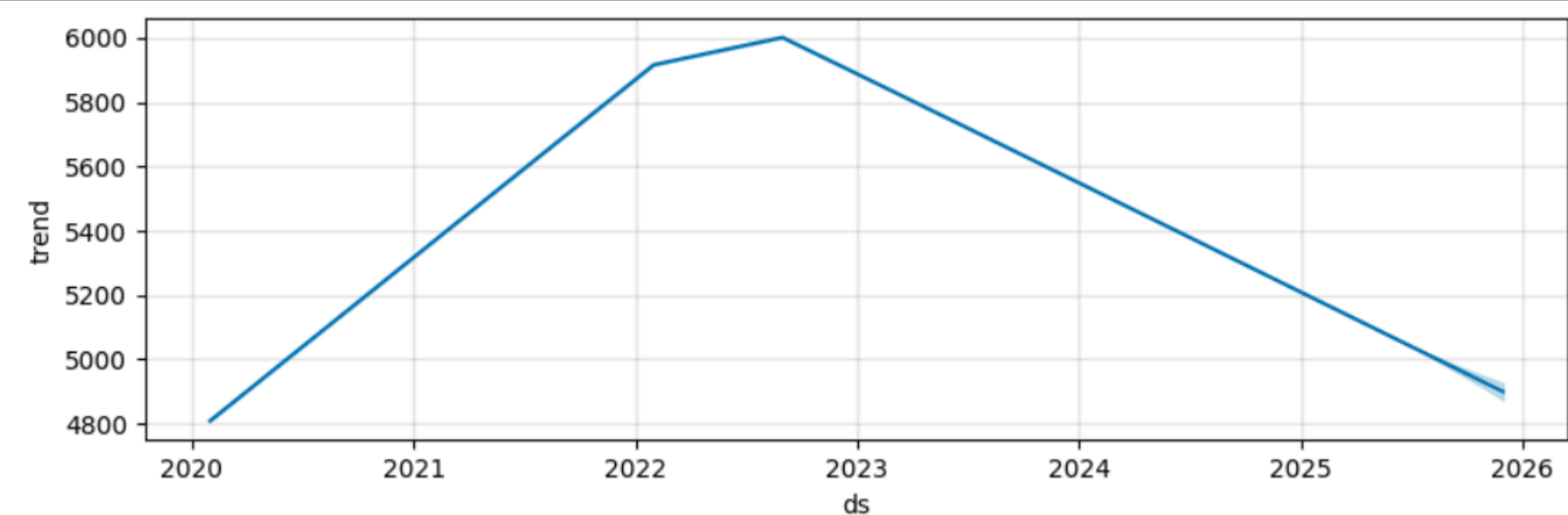
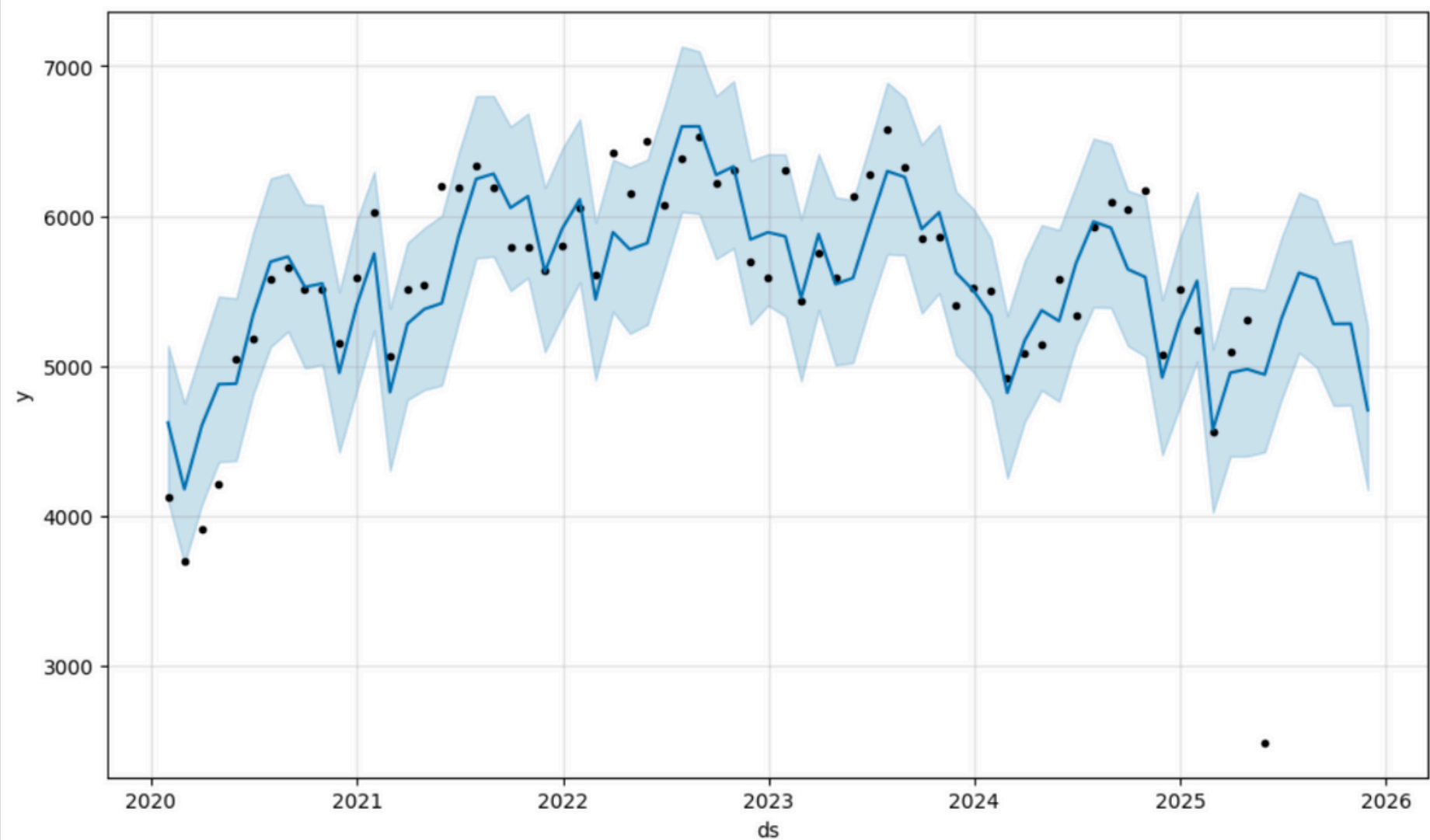
Prophet module for time series analysis

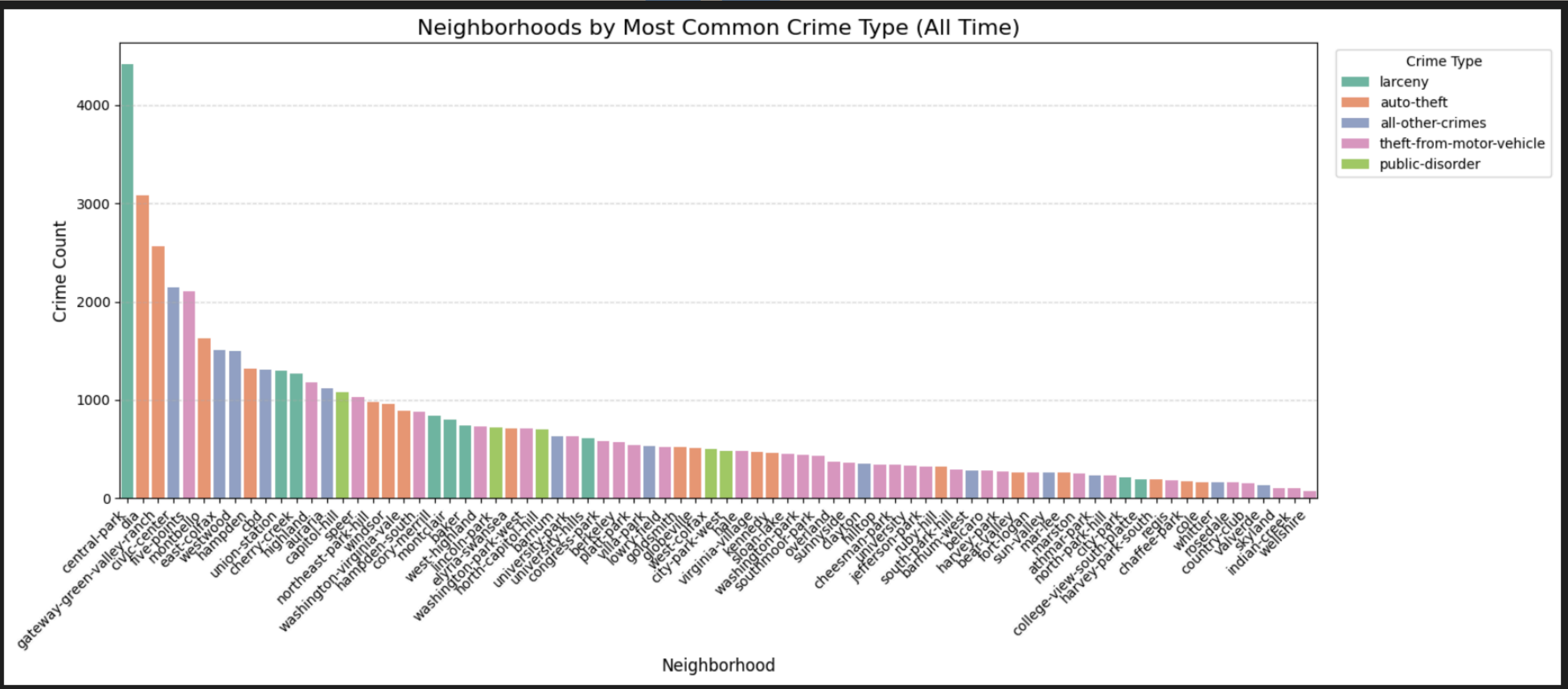


- What is Prophet?
- Open-source forecasting tool developed by Facebook
- Based on an additive model where a time series is decomposed into trend, seasonality, holiday effects, and noise
- Handles non-linear trends
- Robust to missing data, outliers, and sudden trend changes (changepoints)
- Key Features
 - Trend modeling: Automatically detects and models trend changes (changepoints) in the data
 - Seasonality: Models multiple seasonal patterns using Fourier series
- Why Use Prophet?
 - Flexible and interpretable
 - Efficient for large datasets and quick to train
 - Tolerant of missing values, outliers, and abrupt changes

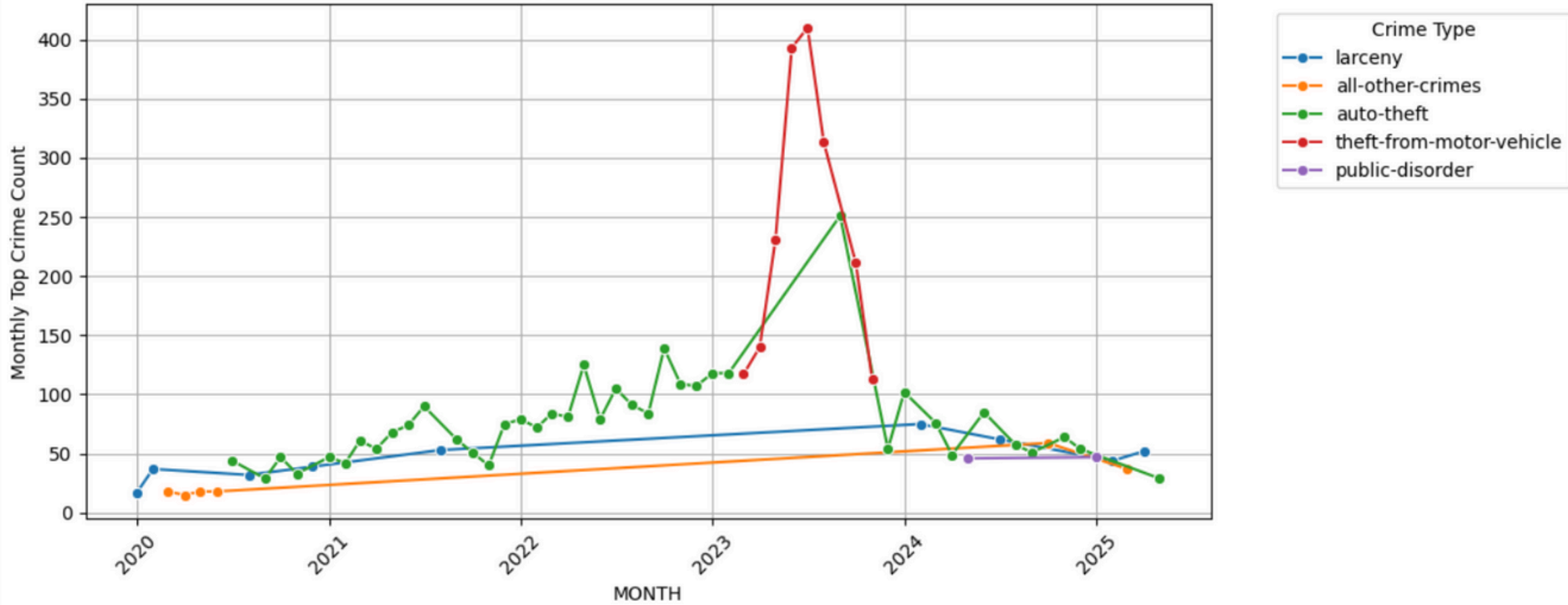


Crime Forecast for the Next 6 Months

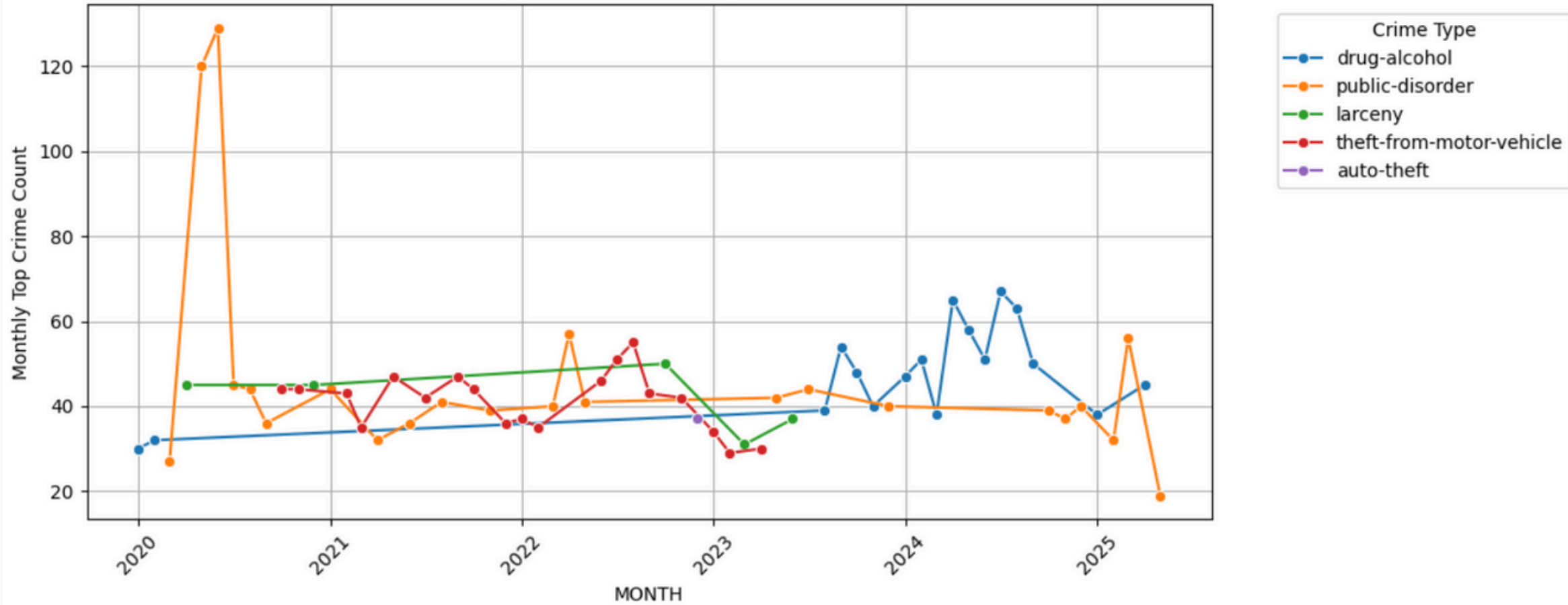


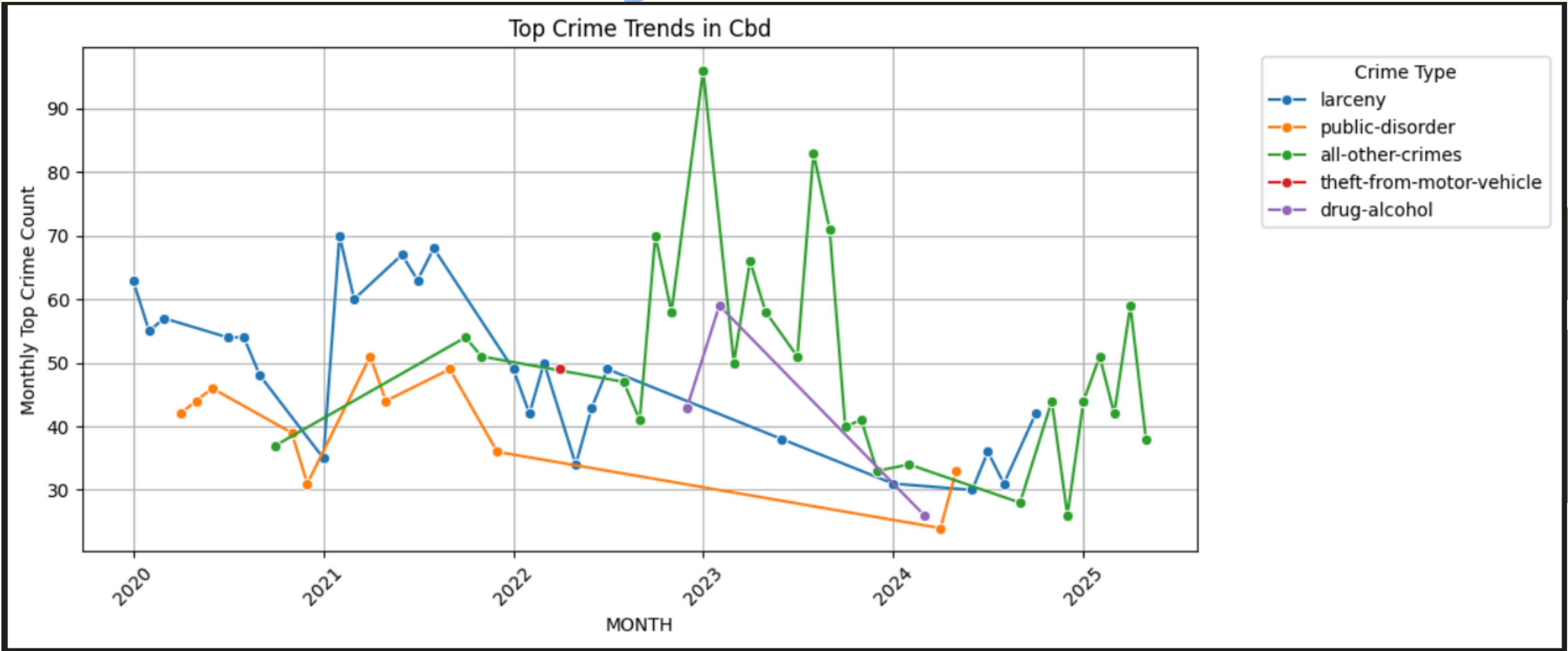


Top Crime Trends in Dia



Top Crime Trends in Capitol-Hill





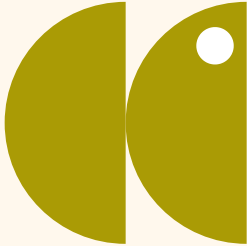


Key Findings

- Clear relationships found between neighborhood demographics and equity scores.
 - After school program counts play a minimal role in understanding equity scores.
 - No obvious relationships were found when analyzing neighborhood level data and crime rates.
 - Hard to predict crime rates given other neighborhood features.
 - Crime rates were high during pandemic years 2020-2022
 - Crime rates have started to decline since 2023
- 



Recommendations



STOP Reactive Deployment

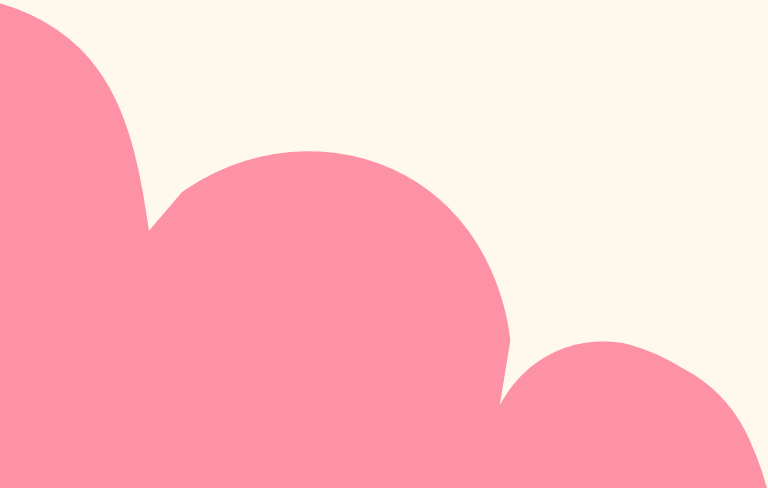
Don't wait for crime spikes to place programs

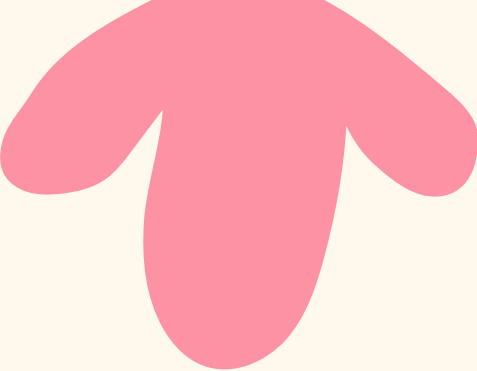
START Proactive Targeting

High poverty + Moderate crime =
Priority intervention

SCALE Comprehensive Approach

Programs + Economic opportunity +
Education



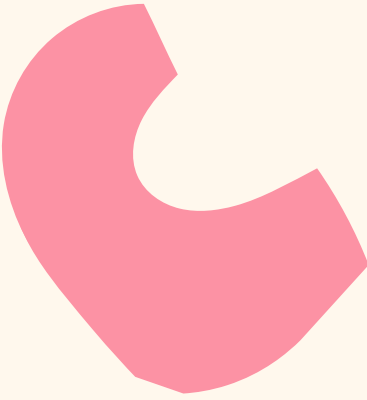


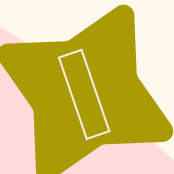
Conclusion

Bottom Line: Data proves Denver needs fundamental strategy change

Key Takeaway: Evidence-based policy beats intuition

Impact: Safer communities through smarter investments





Thank You!

Website:

https://github.com/uelski/4447_Final_Project